

Erdős Problem #25

EP-25

L1 - TRACTABLE

OPEN

Number Theory

ORIGINAL PROBLEM STATEMENT

Let $n_1 < n_2 < \dots$ be an arbitrary sequence of integers, each with an associated residue class $a_i \pmod{n_i}$. Let A be the set of integers n such that for every i either $n < n_i$ or $n \not\equiv a_i \pmod{n_i}$. Must the logarithmic density of A exist?

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Archive: <https://www.unsignedmath.com/problems?difficulty=1>
Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>